

The First Positive Coefficient in Integer Thue Morse Pressure

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Abstract

For the doubling map and the potential $\log \cos^2 \pi(x - c)$, the integer-order pressure has a missing Taylor coefficient at degree $2m$. We prove that the next coefficient, at degree $2m + 2$, is strictly positive for every integer $m \geq 2$. This answers a specific open question in the repository's integer-pressure report. A finite Bernoulli eigenbasis formula proves that the quadratic response of the normalized Perron eigenfunction is positive at every noninteger point. An independent absolutely convergent inverse-branch formula gives an explicit positive lower bound and a large-order expansion with an exponentially smaller error. Exact rational checks accompany the proofs.

1 The question and the result

Let $T(x) = 2x \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and set

$$g_c(x) = \cos^2 \pi(x - c), \quad p_m(c) = P(m \log g_c), \quad m \in \mathbb{N}.$$

We use natural logarithms and the transfer operator defined by a sum, not an average:

$$(\mathcal{L}_{m,c}f)(x) = \sum_{Ty=x} g_c(y)^m f(y). \quad (1)$$

Thus $p_m(c) = \log \rho_m(c)$ near $c = 0$, where ρ_m is the simple dominant eigenvalue of the finite Fourier model described below. Equivalently, if $\mathcal{D}_N$ is the family of dyadic intervals of length 2^{-N} , then

$$p_m(c) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \sum_{I \in \mathcal{D}_N} \sup_{x \in I} \prod_{j=0}^{N-1} g_c(T^j x)^m.$$

The finite model and the identity with pressure are developed in the repository report [1]; a local derivation of the inputs needed here is included in Section 2.

Write

$$R_m = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-2m} = \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}}, \quad C_m = [t^{2m+2}] p_m(t/\pi). \quad (2)$$

The source proves $[t^{2m}] p_m(t/\pi) = 0$ and asks whether $C_m > 0$ for every $m \geq 2$, with exact checks only through $m = 6$. That question remains present in the version inspected for this note, identified precisely in Section 7.

For each half-integer w , define

$$S(w) = \sum_{j \geq 1} \tan \frac{\pi w}{2^j}, \quad U(w) = \sum_{j \geq 1} \tan^2 \frac{\pi w}{2^j}. \quad (3)$$

There are no poles, and both series converge absolutely.

Theorem 1.1. *For every integer $m \geq 2$, the series*

$$B_m = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-2m} (2m^2 S(w)^2 - mU(w)) \quad (4)$$

converges absolutely, and

$$C_m = B_m + \frac{2m}{3} R_m - \frac{m}{m+1} R_{m+1} > 0. \quad (5)$$

In fact,

$$B_m > \frac{2m}{\pi^{2m}} \left(4^m (2m-1) - 16 \left(\frac{2}{3} \right)^{2m-4} \right) > 0. \quad (6)$$

The negative lower even coefficients, the missing coefficient, and the new result together give the local sign pattern

$$p_m(t/\pi) = -2m \sum_{k=1}^{m-1} \frac{(2^{2k}-1)\zeta(2k)}{k\pi^{2k}} t^{2k} + C_m t^{2m+2} + O_m(t^{2m+4}), \quad C_m > 0. \quad (7)$$

The sign statement is uniform over all integer orders, although the local analytic remainder here is for each fixed m .

2 Local transfer data and the feedback identity

The space

$$\mathcal{E}_m = \text{span}_{\mathbb{C}} \{ e^{2\pi i r x} : 1-m \leq r \leq m-1 \}$$

is invariant under (1), because its matrix, indexed by $k, r \in \{1-m, \dots, m-1\}$, has entries

$$A_m(c)_{k,r} = 2 \cdot 4^{-m} \binom{2m}{m+2k-r} e^{-2\pi i (2k-r)c}. \quad (8)$$

A binomial coefficient outside its range is zero.

For completeness, the entire atomic spectrum can be seen directly. For $0 \leq j \leq 2m-2$, put

$$\phi_j(x) = \frac{\sin^{2m}(\pi x)}{\pi^{2m}} \sum_{k \in \mathbb{Z}} (x+k)^{j-2m},$$

with removable values at integers. The identity

$$\sum_{k \in \mathbb{Z}} (x+k)^{-p} = \frac{(-1)^{p-2}}{(p-1)!} \left(\frac{d}{dx} \right)^{p-2} (\pi^2 \csc^2 \pi x), \quad p \geq 2,$$

shows that $\phi_j \in \mathcal{E}_m$: after multiplication by $\sin^{2m} \pi x$, it is a trigonometric polynomial of degree at most $m-1$. Its Taylor expansion at zero starts with x^j , so the $2m-1$ functions are independent. Splitting the periodization into even and odd integers gives

$$\mathcal{L}_{m,0} \phi_j = 2^{-j} \phi_j.$$

Consequently the eigenvalue 1 is simple and strictly dominant. Its eigenfunction ϕ_0 is strictly positive on the circle and satisfies $\phi_0(0) = 1$. Analytic perturbation gives a real-analytic positive eigenfunction $h_{m,c}$ near zero, normalized by $h_{m,c}(0) = 1$, and a simple dominant eigenvalue $\rho_m(c)$.

These data also identify the local pressure. Positivity and comparison with $h_{m,c}$ give $\mathcal{L}_{m,c}^N 1 \asymp_m \rho_m(c)^N$ uniformly in x . If $F_N(x) = \prod_{j < N} g_c(T^j x)^m$, its trigonometric degree is at most $m(2^N - 1)$. The L^1 Bernstein inequality and the variation bound on each dyadic interval give

$$2^N \int_0^1 F_N \leq \sum_{I \in \mathcal{D}_N} \sup_I F_N \leq (1 + 2\pi m) 2^N \int_0^1 F_N.$$

Since $2^N \int F_N = \int \mathcal{L}_{m,c}^N 1$, taking logarithmic growth rates yields $p_m(c) = \log \rho_m(c)$. This is the local part of the more extensive finite-model result in [1].

Set $t = \pi c$, $a = \tan t$, and normalize the transfer operator by its fixed-point weight:

$$(V_a f)(x) = \sum_{Ty=x} (\cos \pi y + a \sin \pi y)^{2m} f(y) = \frac{\mathcal{L}_{m,c} f(x)}{\cos^{2m} t}. \quad (9)$$

Let $\lambda(a)$ and h_a be its analytic eigenvalue and eigenfunction, with $h_a(0) = 1$. The two preimages of zero give the exact identity

$$\lambda(a) = 1 + a^{2m} h_a(1/2). \quad (10)$$

Reflection implies $h_{-a}(1/2) = h_a(1/2)$. Thus

$$h_a(1/2) = R_m + B_m a^2 + O_m(a^4),$$

where we will establish the series (4). As $a = t + O(t^3)$, the quadratic coefficient is unchanged when expressed in t . Now

$$p_m(t/\pi) = 2m \log \cos t + \log \lambda(\tan t).$$

The cosine product gives

$$\log \cos t = - \sum_{k \geq 1} \frac{(2^{2k} - 1) \zeta(2k)}{k \pi^{2k}} t^{2k}.$$

Together with $\tan^{2m} t = t^{2m} (1 + \frac{2m}{3} t^2 + O(t^4))$, this proves (5) as an identity. The square term in the logarithm starts at degree $4m > 2m + 2$ for $m \geq 2$. The coefficient at degree $2m$ cancels against R_m , proving the asserted missing term as well.

3 A finite Bernoulli formula and pointwise positivity

The sign also admits a finite proof that strengthens the result to the whole quadratic eigenfunction response. Define

$$H_p(x) = \sum_{n \in \mathbb{Z}} (\pi(x+n))^{-p} \quad (p \geq 2), \quad H_1(x) = \cot \pi x, \quad c_j = \frac{2\zeta(2j)}{\pi^{2j}},$$

and use the atomic eigenbasis

$$F_r(x) = \sin^{2m}(\pi x) H_{2m-r}(x), \quad 0 \leq r \leq 2m - 2.$$

Thus $F_r = \pi^r \phi_r$, $\mathcal{L}_{m,0} F_r = \lambda_r F_r$ with $\lambda_r = 2^{-r}$, and F_r starts with $(\pi x)^r$ at zero. In particular, normalization at zero fixes the coefficient of F_0 .

Theorem 3.1. *Put*

$$f(\lambda, \mu) = \frac{4 - \lambda(3 - \mu)}{2(1 - \lambda)(1 - \mu)}.$$

For $1 \leq j \leq m - 1$, define the positive rational number

$$\begin{aligned} b_{m,j} = 4m^2 \sum_{k=1}^j c_k c_{j+1-k} f(4^{-j}, 2^{-(2k-1)}) \\ - \mathbf{1}_{\{j \leq m-2\}} 2m(2j+1) c_{j+1} f(4^{-j}, 2^{-(2j+1)}). \end{aligned} \quad (11)$$

Then

$$[a^2]h_a(x) = \sum_{j=1}^{m-1} b_{m,j} F_{2j}(x) > 0 \quad (x \notin \mathbb{Z}), \quad (12)$$

while this coefficient is zero at integers. In particular,

$$B_m = \sum_{j=1}^{m-1} b_{m,j} R_{m-j}. \quad (13)$$

Every term is explicit and rational, since the c_j are rational Bernoulli values. More precisely,

$$b_{m,j} \geq 2mf(4^{-j}, 2^{-(2j+1)})(2m(2j+3) - (2j+1))c_{j+1} > 0. \quad (14)$$

Proof. The cotangent product identity is

$$H_1 H_p = H_{p+1} - \sum_{k=1}^{\lfloor (p-1)/2 \rfloor} c_k H_{p+1-2k}, \quad p \geq 2. \quad (15)$$

To verify it, use $\cot z = z^{-1} - \sum_{k \geq 1} c_k z^{2k-1}$ in the local coordinate $z = \pi x$. All principal parts cancel. When p is even, the possible simple pole cancels between the regular constant $c_{p/2}$ of H_p and the corresponding term of cotangent; when p is odd no simple pole occurs. The difference of the two sides is entire and one-periodic, and tends to zero at both imaginary ends of a fundamental strip, since each H_p , $p \geq 2$, is a derivative of $\csc^2$. Liouville's theorem proves (15).

Let $A = \pi^{-1}d/dx$ in the F basis. Differentiation and (15) give

$$AF_r = rF_{r-1} - 2m \sum_{k=1}^{\lfloor (2m-r-1)/2 \rfloor} c_k F_{r+2k-1}. \quad (16)$$

The first summand is absent at $r = 0$. Thus $A_{s,r} = r$ if $s = r - 1$, and $A_{s,r} = -2mc_k$ if $s = r + 2k - 1$; all other entries vanish.

For translation $\mathcal{T}_u f(x) = f(x + u)$ and $L = \text{diag}(\lambda_r)$, the normalized operator satisfies

$$V_a = (1 + a^2)^m \mathcal{T}_{-2 \arctan(a)/\pi} L \mathcal{T}_{\arctan(a)/\pi}.$$

Consequently $V_a = L + aV_1 + a^2V_2 + O(a^3)$, where

$$V_1 = LA - 2AL, \quad V_2 = \frac{1}{2}(LA^2 - 4ALA + 4A^2L) + mL.$$

Expand $h_a = e_0 + ah_1 + a^2h_2 + O(a^3)$ in the F basis. Normalization makes $(h_1)_0 = (h_2)_0 = 0$, and (10) shows that the eigenvalue has no linear or quadratic term for $m \geq 2$. Solving the first two equations gives

$$(h_1)_s = A_{s,0} \frac{\lambda_s - 2}{1 - \lambda_s}, \quad (h_2)_r = \sum_{s \geq 1} A_{r,s} A_{s,0} f(\lambda_r, \lambda_s) \quad (r \geq 1).$$

For the second identity, the coefficient before division by $1 - \lambda_r$ is

$$\frac{(\lambda_r - 2\lambda_s)(\lambda_s - 2)}{1 - \lambda_s} + \frac{\lambda_r}{2} - 2\lambda_s + 2 = \frac{4 - \lambda_r(3 - \lambda_s)}{2(1 - \lambda_s)}.$$

Only odd s contribute, and only even r remain. At $r = 2j$, the upward entries in (16) yield the sum in (11). The sole possible downward entry has $s = 2j + 1$, yielding its negative term. That entry is absent when $j = m - 1$.

For $0 < \lambda, \mu < 1$, we have $f > 0$ and

$$\partial_\mu f(\lambda, \mu) = \frac{2 - \lambda}{(1 - \lambda)(1 - \mu)^2} > 0.$$

All positive terms in (11) have a larger second argument than $2^{-(2j+1)}$. Euler's cotangent convolution,

$$\sum_{k=1}^j c_k c_{j+1-k} = (2j + 3)c_{j+1}, \quad (17)$$

is obtained by comparing the coefficient of z^{2j} in $(\cot z)' = -1 - \cot^2 z$. Monotonicity and (17) prove (14); if the negative term is absent, subtracting it only weakens the bound. Finally, each $F_{2j}(x)$ is positive away from integers, because H_{2m-2j} is a sum of positive even powers. This proves (12), and evaluation at $1/2$ gives (13). $\square$

The pointwise conclusion concerns the quadratic Taylor coefficient. A linear phase response can be present at a general x , so it does not assert a local phase minimum everywhere. Formula (13), together with (5), is a finite Bernoulli expression for the requested coefficient. The orbit representation below supplies a complementary geometric proof, the explicit bound (6), and the large-order asymptotic.

4 The inverse branch formula

Lemma 4.1. *The quadratic coefficient $[a^2]h_a(1/2)$ is the absolutely convergent series (4).*

Proof. Put $F_N(a, x) = V_a^N 1(x)$. Evaluation at zero gives

$$F_{N+1}(a, 0) = F_N(a, 0) + a^{2m} F_N(a, 1/2),$$

so $F_N(a, 0) = 1 + O(a^{2m})$ for every finite N . Therefore

$$[a^2]F_N(a, 1/2) = [a^2] \frac{F_N(a, 1/2)}{F_N(a, 0)}, \quad m \geq 2. \quad (18)$$

The ratio on the right converges to $h_a(1/2)$ uniformly on a sufficiently small complex disk around zero. Indeed, analytic spectral projection separates $\lambda(a)$ from all remaining eigenvalues by a uniform modulus ratio less than one. The leading projection of 1 is nonzero at $a = 0$, as $F_N(0, 0) = 1$, and remains nonzero after shrinking the disk. A separating-contour bound makes the spectral remainder exponentially smaller than $\lambda(a)^N$, uniformly on that disk. Thus the denominator is nonzero for all sufficiently large N , and Cauchy's coefficient formula yields

$$[a^2]h_a(1/2) = \lim_{N \rightarrow \infty} [a^2]F_N(a, 1/2). \quad (19)$$

Use centered inverse branches

$$W_N = \{w \in \mathbb{Z} + \frac{1}{2} : -2^{N-1} < w < 2^{N-1}\}.$$

Changing the conventional branch representative by a multiple of 2^N preserves every tangent and every even cosine power. The unperturbed product telescopes to

$$\prod_{j=1}^N \cos^{2m} \frac{\pi w}{2^j} = \left(2^N \sin \frac{\pi w}{2^N} \right)^{-2m}, \quad w \in W_N,$$

because $\sin \pi w = \pm 1$. Write S_N, U_N for the sums in (3) truncated at $j = N$. The quadratic coefficient of the relative product $\prod_{j=1}^N (1 + a \tan(\pi w/2^j))^{2m}$ is exactly $2m^2 S_N(w)^2 - m U_N(w)$. Hence

$$[a^2]F_N(a, 1/2) = \sum_{w \in W_N} \left(2^N \sin \frac{\pi w}{2^N} \right)^{-2m} (2m^2 S_N(w)^2 - m U_N(w)). \quad (20)$$

We record uniform bounds to justify the limit, including $m = 2$. By symmetry assume $w \geq 1/2$, and put $J = \lceil \log_2(2w) \rceil$, so $2w \leq 2^J < 4w$. Since w is a half-integer, its scaled distance from any tangent pole is at least $2^{-(j+1)}$. Thus

$$\left| \tan \frac{\pi w}{2^j} \right| \leq \cot \frac{\pi}{2^{j+1}} < \frac{2^{j+1}}{\pi}.$$

For $j \geq J+1$, the argument lies in $[0, \pi/4]$, and convexity gives the sharper bound $|\tan(\pi w/2^j)| \leq 4w/2^j$. Summing the two ranges proves

$$\sum_{j \geq 1} \left| \tan \frac{\pi w}{2^j} \right| < \frac{4}{\pi} 2^J + \frac{4w}{2^J} \leq \left(\frac{16}{\pi} + 4 \right) w < 10w, \quad (21)$$

$$U(w) < \frac{16}{3\pi^2} 4^J + \frac{16w^2}{3 \cdot 4^J} \leq \left(\frac{256}{3\pi^2} + \frac{16}{3} \right) w^2 < 16w^2. \quad (22)$$

In the second line we used $4/3 \leq 16w^2/3$ and $\pi^2 > 8$.

For $w \in W_N$, the chord bound for sine gives

$$\left| 2^N \sin \frac{\pi w}{2^N} \right|^{-2m} \leq (2|w|)^{-2m}.$$

Consequently the absolute summand in (20) is bounded by a constant depending on m times $|w|^{2-2m}$. This is summable on the half-integer lattice for every $m \geq 2$. Each fixed summand has the limit claimed in (4). Dominated convergence, followed by (19), proves the lemma. $\square$

5 Two branches force positivity

For $w = 1/2$, all tangent terms are positive and the first equals one. Therefore

$$S(1/2)^2 \geq U(1/2) \geq 1.$$

The two branches $w = \pm 1/2$ contribute at least $2(2/\pi)^{2m} m(2m-1)$ to (4). For all other branches we discard the nonnegative square term and use (22), obtaining

$$B_m \geq \frac{2m}{\pi^{2m}} \left(4^m (2m-1) - 16 \sum_{k \geq 1} (k + \frac{1}{2})^{-(2m-2)} \right).$$

Strict convexity of x^{-2} gives the midpoint estimate

$$\sum_{k \geq 1} (k + \frac{1}{2})^{-2} < \int_1^\infty x^{-2} dx = 1.$$

Since $k + 1/2 \geq 3/2$,

$$\sum_{k \geq 1} (k + \frac{1}{2})^{-(2m-2)} < \left(\frac{2}{3}\right)^{2m-4}.$$

This proves (6); its bracket is positive at $m = 2$ and increases thereafter.

Finally, (2) implies $R_{m+1} \leq (4/\pi^2)R_m$, so

$$\frac{2m}{3}R_m - \frac{m}{m+1}R_{m+1} \geq mR_m \left(\frac{2}{3} - \frac{4}{(m+1)\pi^2}\right) > 0.$$

Combining this with $B_m > 0$ proves Theorem 1.1.

6 Large orders and exact evaluation

Set

$$S_0 = S(1/2), \quad U_0 = U(1/2).$$

Corollary 6.1. *As $m \rightarrow \infty$ through integers,*

$$\begin{aligned} C_m = & \left(\frac{2}{\pi}\right)^{2m} \left(4m^2 S_0^2 + m \left(\frac{4}{3} - 2U_0\right) - \frac{8m}{(m+1)\pi^2}\right) \\ & + O\left(m^2 \left(\frac{2}{3\pi}\right)^{2m}\right). \end{aligned} \tag{23}$$

In particular, $C_m \sim 4S_0^2 m^2 (2/\pi)^{2m}$.

Proof. The $\pm 1/2$ terms give the displayed main expression exactly. From (21)–(22), the remaining part of B_m has absolute value at most

$$2m(200m + 16)\pi^{-2m} \left(\frac{2}{3}\right)^{2m-4} = O\left(m^2 \left(\frac{2}{3\pi}\right)^{2m}\right).$$

The same midpoint estimate bounds the tails of R_m and R_{m+1} by smaller quantities of this order. All constants in this argument can be chosen independent of $m \geq 2$. $\square$

The coefficients are rational and can be computed without truncating the orbit sum. In (8), let $A = A_m(0)$, $j = 2k - r$, and introduce the real rational matrices

$$D_{k,r} = -2jA_{k,r}, \quad E_{k,r} = (m - 2j^2)A_{k,r}.$$

The normalized matrix of V_a is $A + iaD + a^2E + O(a^3)$. Solve

$$\begin{aligned} (I - A)v_0 &= 0, & \mathbf{1}^T v_0 &= 1, \\ (I - A)u_1 &= Dv_0, & \mathbf{1}^T u_1 &= 0, \\ (I - A)v_2 &= Ev_0 - Du_1, & \mathbf{1}^T v_2 &= 0. \end{aligned}$$

These systems have unique solutions because the eigenvalue 1 is simple and its left eigenvector is evaluation at zero, namely $\mathbf{1}^T$. With $\ell_r = (-1)^r$, we have $B_m = \ell^T v_2$. Equation (5) then yields C_m exactly.

m	B_m	C_m
2	8	376/45
3	155/21	213/28
4	1105984/205065	16978624/3075975
5	808150982/234389295	9058569913/2578282245
6	30435818208256/15028650946575	57427231423264/27910351757925

The independent checkers verify these systems through $m = 10$, compare them with the finite Bernoulli formula and its coefficientwise positive bounds, and exactly match the three coefficients printed in [1]. A separate high-precision comparison checks finite inverse-tree identities against rational matrix powers. These are regression checks; the all-order proof is the analytic argument above.

7 Source status and further questions

The specific target is the question entitled “The first coefficient after the cancellation” in [1]. We inspected repository commit `6a98f89bac85cc6b4ba5d5c3b63996037d9824e1`, with source blob `1973fd17245864bb8a83e1b1d90416ac032ba7c1`. The source already establishes integer phase analyticity, the atomic spectrum, the missing coefficient, and the low-order calculations. Those results are credited here. The new results relative to that source are the all- m positive sign, the finite Bernoulli formula and pointwise response positivity, the inverse-branch representation, the quantitative bound, and the large-order expansion.

The primary literature comparison is Gohlke–Kesseböhmer–Schindler [2, Theorem 2.7 and following discussion], which proves phase analyticity at order two and discusses general positive-order regularity. The arXiv record checked for this note lists version 1, submitted 26 September 2025. A targeted literature search did not locate the coefficient theorem proved here; this is not an exhaustive priority claim. The arguments have been checked by direct derivation and independent exact computations. They are not externally refereed or formally verified.

Question 1. *Can the reduced denominators of C_m be characterized uniformly from (11)–(13)? The finite formula gives candidate factors from Bernoulli denominators and $1 - 2^{-r}$, but cancellations still need to be understood.*

Question 2. *What is the sign pattern of subsequent coefficients $[t^{2m+2r}]p_m(t/\pi)$ for $r \geq 2$? Higher inverse-branch derivatives introduce products of tangent sums, and the domination used here requires additional analysis at the smallest moment orders.*

Question 3. *Can the local expansion be made uniform in a joint limit $m \rightarrow \infty$, $t \rightarrow 0$? Formula (23) concerns the coefficient sequence; it does not provide a uniform radius or remainder for the pressure itself.*

References

- [1] *Integer Pressure and a Missing Taylor Coefficient: Analytic phase dependence and the sixth-moment optimum for generalized Thue–Morse products.* ProveIt research manuscript, 28 September 2026, current version inspected 1 October 2026.
Versioned repository source
- [2] P. Gohlke, M. Kesseböhmer, and T. I. Schindler, *Generalized Thue–Morse measures: spectral and fractal analysis*, arXiv:2509.22109v1, 2025.
<https://arxiv.org/html/2509.22109v1>